

Mapping the Consumption Gap:
Planned Data Center Capacity versus Electricity Supply in
California

September 21, 2026

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Abstract

Chapter 1

Introduction

Chapter 2

Current Status of Data Centers and the Grid in California

This chapter establishes the baseline that the gap analysis in later chapters is measured against. It answers two questions: what the CAISO grid looks like hour by hour today, and how much electricity California’s existing data centers already use. Every number is computed from the raw files listed in `data/raw/manifest.csv`; the notebook `notebooks/01_baseline_grid.ipynb` reproduces each table and figure.

2.1 Grid baseline: CAISO demand, net load and imports, 2019–2025

Hourly demand, generation by fuel and net interchange for the CAISO balancing authority come from the EIA-930 Hourly Electric Grid Monitor, using EIA’s “Adjusted” series [U.S. Energy Information Administration, 2026a]. Net load is demand minus utility-scale solar and wind. Two definitional points matter. First, EIA-930 demand for CISO includes battery charging and pumping load, whereas CAISO’s own reported demand excludes charging; in 2025 the hourly gap between the two series tracks battery charging almost one for one (slope 0.91, correlation 0.87). Both definitions are kept in the processed table. Second, EIA’s series contains reporting artifacts, chiefly in early 2019, where demand falls implausibly below CAISO’s demand plus charging; those 347 hours are nulled rather than corrected, and annual energy is computed from the mean of the remaining hours. Third, EIA-930 carries no CISO interchange at all, in any of its reported, imputed or adjusted variants, from July 2 to November 3, 2024, and no hydro for parts of 2019 and 2020; those hours are filled from CAISO’s own five-minute imports and hydro series through a per-year linear calibration fitted on the overlapping hours (correlation 0.986 or better; the fill log is `ch1_eia930_fill_log.csv`).

Table 2.1: CAISO annual baseline from EIA-930 (Adjusted series), 2019–2025. Storage charging from CAISO’s five-minute fuel mix. Import share is net imports divided by demand.

Year	Demand (TWh)	Peak (GW)	Load factor	Peak net load (GW)	Min net load (GW)	Solar (TWh)	Wind (TWh)	Storage charging (TWh)	Net imports (TWh)	Imports (% of demand)
2019	215.8	43.8	0.56	40.5	5.2	27.7	15.9	0.1	53.8	25
2020	218.0	46.6	0.53	44.0	5.6	28.6	14.9	0.1	59.4	27
2021	219.2	43.6	0.57	40.5	3.0	32.8	18.3	0.9	54.5	25
2022	223.7	51.1	0.50	47.5	1.9	35.8	17.4	1.7	48.7	22
2023	218.2	44.0	0.57	41.8	1.7	37.2	16.4	4.4	28.8	13
2024	224.1	47.6	0.54	42.4	0.3	44.8	20.1	9.5	31.7	14
2025	224.1	43.9	0.58	39.8	-3.3	49.7	19.8	14.2	36.1	16

Table 2.1 shows a grid whose annual demand has been flat since 2019 while its composition changed. Demand ranged between 216 and 224 TWh in every year. Utility-scale solar output rose from 27.7 TWh to 49.7 TWh and the combined solar and wind share of demand went from a fifth to nearly a third. Net imports fell from a quarter of demand in 2019 to 13 percent in 2023 and 14 to 16 percent in 2024 and 2025. Battery charging, negligible in 2019, reached 14 TWh in 2025, and battery discharge peaked above 10 GW (Table 2.5). The hourly demand record remains 51.1 GW, set on September 6, 2022 at 18:00.

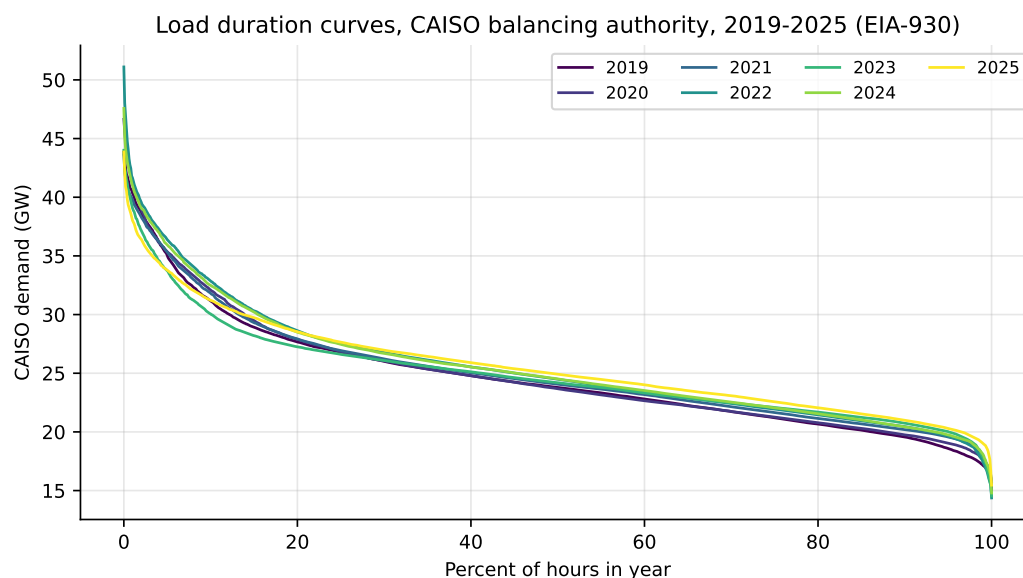


Figure 2.1: Load duration curves, CAISO, 2019–2025. Source: EIA-930.

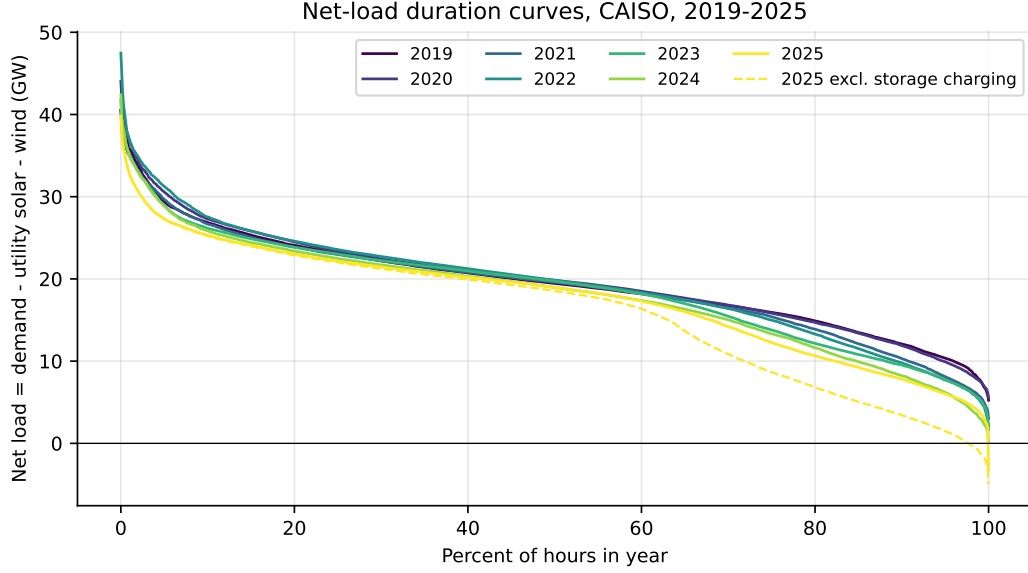


Figure 2.2: Net-load duration curves, CAISO, 2019–2025. The dashed line removes battery charging from 2025 demand. Source: EIA-930, CAISO fuel mix.

Figures 2.1 and 2.2 make the contrast visible. The load duration curves are nearly indistinguishable across years. The net-load duration curves diverge at the low end: by 2024 and 2025 the lowest-net-load hours approach zero, and once battery charging, which is itself a response to the surplus, is removed from demand, utility solar and wind alone exceeded end-use demand in 80 hours of 2024 and 205 hours of 2025. This midday surplus is the physical basis for the flexibility question in Chapter 5.

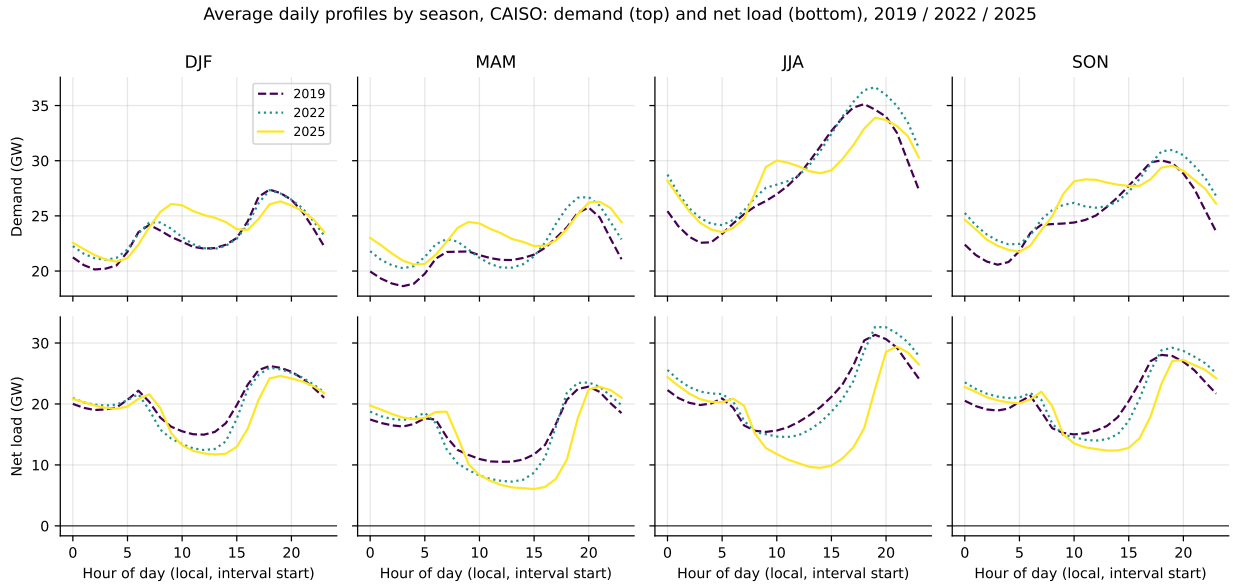


Figure 2.3: Average daily profiles by season: demand (top) and net load (bottom), 2019, 2022 and 2025. Source: EIA-930.

The seasonal profiles in Figure 2.3 show the mechanism. Demand profiles are similar across years, with a modest midday rise in 2025 that reflects storage charging. Net-load profiles deepen at midday and steepen in the evening: in spring the average midday net load fell from about 10.5 GW in 2019 to about 6 GW in 2025 while the 20:00 value stayed near 22 GW, and in summer the midday trough fell from about 15 GW to under 10 GW.

Table 2.2: The 100 highest net-load hours of each year: when they occur and what serves them. Source: EIA-930.

Year	Net load range (GW)	Distinct days	Jul–Sep (%)	17:00–21:00 (%)	Median start	Mean solar (GW)	Mean imports (GW)	Mean gas (GW)
2019	35.6–40.5	34	90	99	18:00	0.9	8.5	18.5
2020	36.4–44.0	30	92	86	18:00	1.2	7.4	22.6
2021	34.7–40.5	42	92	100	19:00	0.6	6.3	21.0
2022	36.6–47.5	27	95	82	19:00	1.2	7.4	22.5
2023	35.0–41.8	35	96	93	20:00	0.6	3.6	21.8
2024	34.7–42.4	37	91	79	20:00	0.9	4.7	20.9
2025	32.3–39.8	33	98	78	20:00	0.7	4.6	17.3

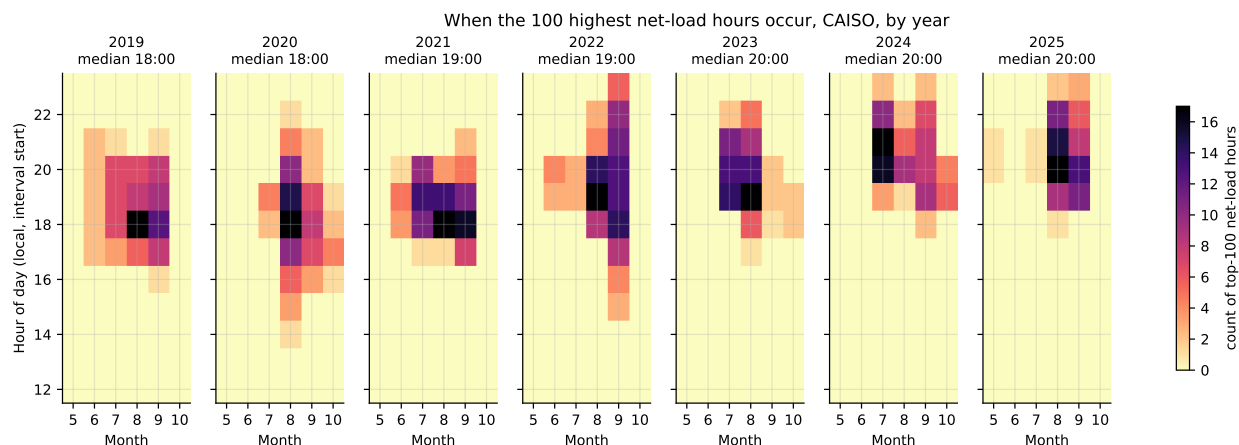


Figure 2.4: Month and hour of the 100 highest net-load hours, by year. Source: EIA-930.

Table 2.2 and Figure 2.4 identify the hours that set capacity needs. In every year at least 90 percent of the top 100 net-load hours fall in July through September and at least 78 percent between 17:00 and 21:00. Their median start time moved from 18:00 in 2019 and 2020 to 19:00 in 2021 and 2022 and to 20:00 from 2023 onward, the signature of solar pushing the net peak past sunset and of storage shaving the early evening. Solar contributes between 0.6 and 1.2 GW on average in these hours; gas contributes 17 to 23 GW and imports 4 to 9 GW. The implication for a flat load such as a data center is direct: each megawatt adds one for one to hours that solar does not serve, and the relevant capacity resources are storage, gas and imports.

2.2 Day-ahead prices and negative-price hours

Hourly day-ahead prices at the three investor-owned utility default load aggregation points come from CAISO OASIS [California Independent System Operator, 2026b]. OASIS retains roughly 39

months of data, so hourly prices run from July 2023; 2024 and 2025 are complete years and 2026 covers January 1 to September 22. EIA's daily ICE hub prices at NP15 and SP15 extend to 2015 but are daily block averages [U.S. Energy Information Administration, 2026b].

Table 2.3: Day-ahead LMP statistics at CAISO default load aggregation points. Source: CAISO OASIS PRC_LMP, DAM.

Node	Year	Hours	Mean	Median	P5	P95	Min	Hours < \$0	Hours $\leq$ \$5	Negative hours 10:00–16:00 (%)
(\$/MWh)										
PG&E DLAP	2024	8784	43.0	40.1	9.0	76.2	-33.1	190	327	88
PG&E DLAP	2025	8760	40.6	42.4	8.7	63.8	-36.8	218	337	94
PG&E DLAP	2026*	6359	31.5	31.9	0.4	60.1	-14.2	294	480	87
SCE DLAP	2024	8784	34.0	37.0	-20.0	67.7	-66.6	1131	1369	77
SCE DLAP	2025	8760	34.4	40.2	-8.3	60.5	-40.5	877	1109	78
SCE DLAP	2026*	6359	28.7	30.3	-9.0	64.8	-17.9	890	1123	72
SDG&E DLAP	2024	8784	36.2	38.2	-17.3	72.5	-63.1	989	1227	77
SDG&E DLAP	2025	8760	37.5	42.4	-6.8	65.8	-40.1	748	955	80
SDG&E DLAP	2026*	6359	32.3	33.1	-7.2	70.3	-17.2	706	950	74

* partial year: 2026 covers January 1 to September 22.

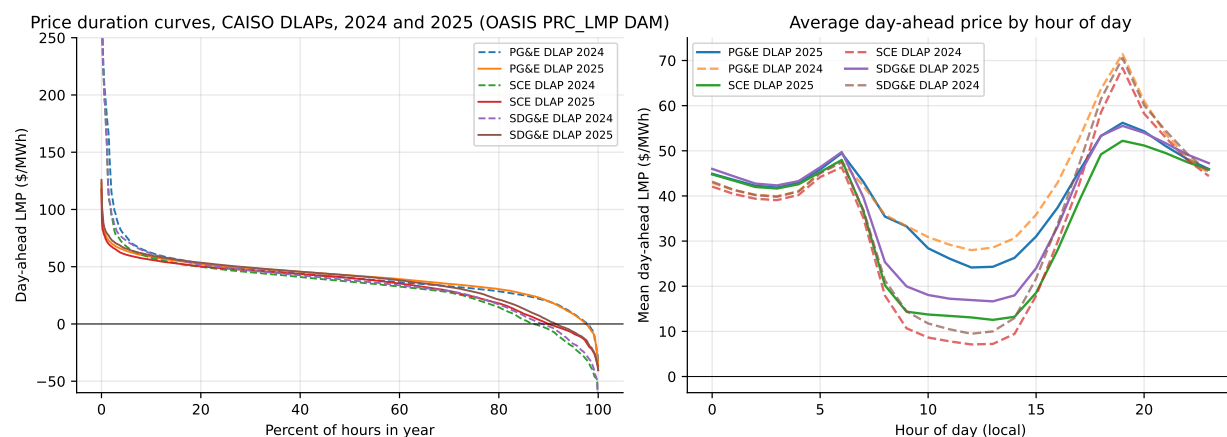


Figure 2.5: Price duration curves (left) and average price by hour of day (right), 2024 and 2025. Source: CAISO OASIS.

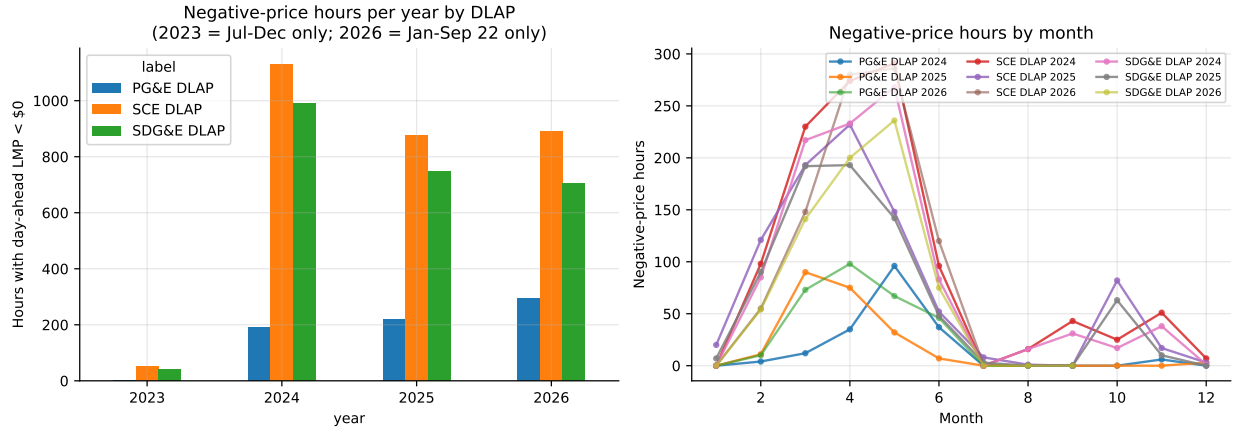


Figure 2.6: Negative-price hours per year (left) and by month (right) at the three DLAPs. Source: CAISO OASIS.

Negative day-ahead prices are now routine in Southern California (Table 2.3, Figures 2.5 and 2.6). The SCE aggregation point cleared below zero in 1,131 hours of 2024 and 877 hours of 2025, and in 890 hours of 2026 through September 22; SDG&E's point saw 989 and 748 hours; PG&E's 190 and 218. Between 72 and 94 percent of those hours fall between 10:00 and 16:00, concentrated in February through June, and the average price by hour of day drops to \$7–\$30 per MWh at midday against \$50–\$70 in the evening peak. The daily ICE block prices show almost no negative days, which is why the hourly series is required. This is the California-specific fact that Chapter 5 uses: a data center able to place consumption in those hours faces a zero or negative marginal energy price, and one that cannot adds to the evening hours where prices and net load are highest.

2.3 Hourly carbon intensity

CAISO's Today's Outlook reports five-minute CO₂ emissions by source, including imports at CAISO's default factor [California Independent System Operator, 2026c]. Intensity is computed as total CO₂ divided by CAISO demand, energy-weighted per year. Because CAISO nets export emissions against imports, net CO₂ is negative in export-heavy hours (374 hours in 2024, 425 in 2025); intensity is floored at zero in those hours rather than dropping them. The EPA eGRID output emission rate for the CAMX subregion is the annual reference factor [U.S. Environmental Protection Agency, 2025].

Table 2.4: Carbon intensity of CAISO demand (load-based, including imports) and the eGRID CAMX annual output rate. Source: CAISO Today’s Outlook, eGRID 2023.

Year	Energy-weighted (g/kWh)	P5 (g/kWh)	P95 (g/kWh)	CO ₂ (Mt)	Imports share (%)	Gas share (%)
2019	257	109	361	55.9	42	53
2020	277	132	368	60.4	42	54
2021	274	125	361	59.7	37	59
2022	264	108	355	58.1	34	63
2023	242	85	344	51.1	21	75
2024	201	7	322	42.5	27	68
2025	188	1	319	38.8	35	61
eGRID 2023 CAMX	198	—	—	47.3	—	—

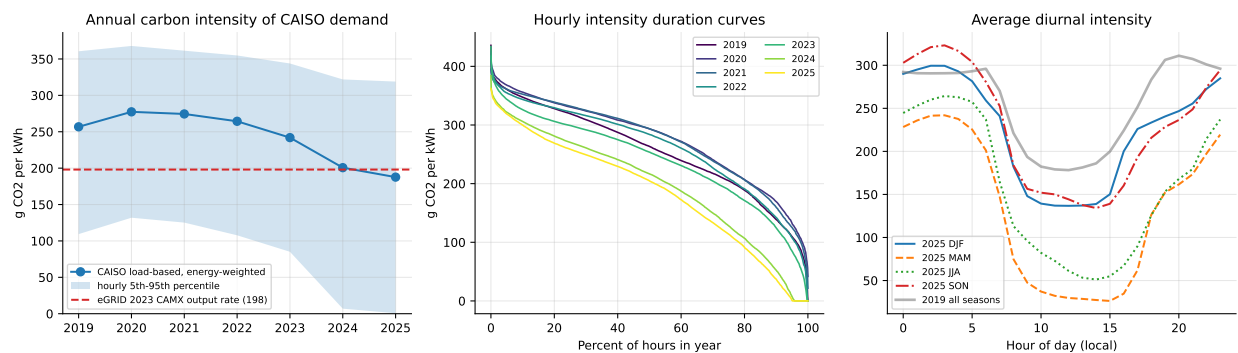


Figure 2.7: Annual, duration-curve and diurnal views of CAISO carbon intensity. Source: CAISO Today’s Outlook, eGRID 2023.

Table 2.5: CAISO battery fleet energy and peak power from the five-minute fuel mix.

Year	Discharge (TWh)	Charge (TWh)	Peak discharge (GW)	Peak charge (GW)
2019	0.05	0.08	0.1	0.1
2020	0.06	0.08	0.2	0.2
2021	0.83	0.92	1.4	1.2
2022	2.73	1.70	3.0	2.1
2023	4.34	4.45	4.8	4.3
2024	8.20	9.45	7.8	7.2
2025	12.29	14.21	10.7	9.3

Load-based intensity fell from 257 g/kWh in 2019 to 188 g/kWh in 2025 (Table 2.4, Figure 2.7). The hourly spread widened: in 2025 the fifth percentile was below 1 g/kWh, the ninety-fifth 319 g/kWh, and the average midday hour was under 30 g/kWh in spring and about 50 g/kWh in summer against roughly 240 to 300 g/kWh at night. eGRID’s 2023 CAMX output rate of 198 g/kWh is lower than the CAISO load-based figure for 2023 (242 g/kWh) because it is generation-based, excludes the default import factor and covers the whole CAMX subregion; both measures show the same decline. For a flat load, the annual factor is the right one; for a flexible load, the diurnal difference of roughly 4 to 1 is the relevant number.

2.4 How much of the state’s power do data centers use

No agency meters California’s data center consumption. Four independent estimates are therefore presented against California retail sales of 245.7 TWh in 2024 from EIA-861 (Parts A, C and D, which avoids counting delivery-only sales twice) [U.S. Energy Information Administration, 2025]. No single estimate is chosen; the spread is the finding.

Table 2.6: Existing California data center electricity use, four estimates.

Method	TWh per year	Share of 246 TWh (%)
EPRI 2024 state table (2023)	9.3	3.8
CEC existing peak demand x load factor	7.0–8.3	2.9–3.4
Count-based bottom-up: Kollar-Grady facilities x avg load (Epoch: no CA sites)	7.0–20.7	2.8–8.4
Silicon Valley Power cluster only (lower bound)	2.4–2.8	1.0–1.1

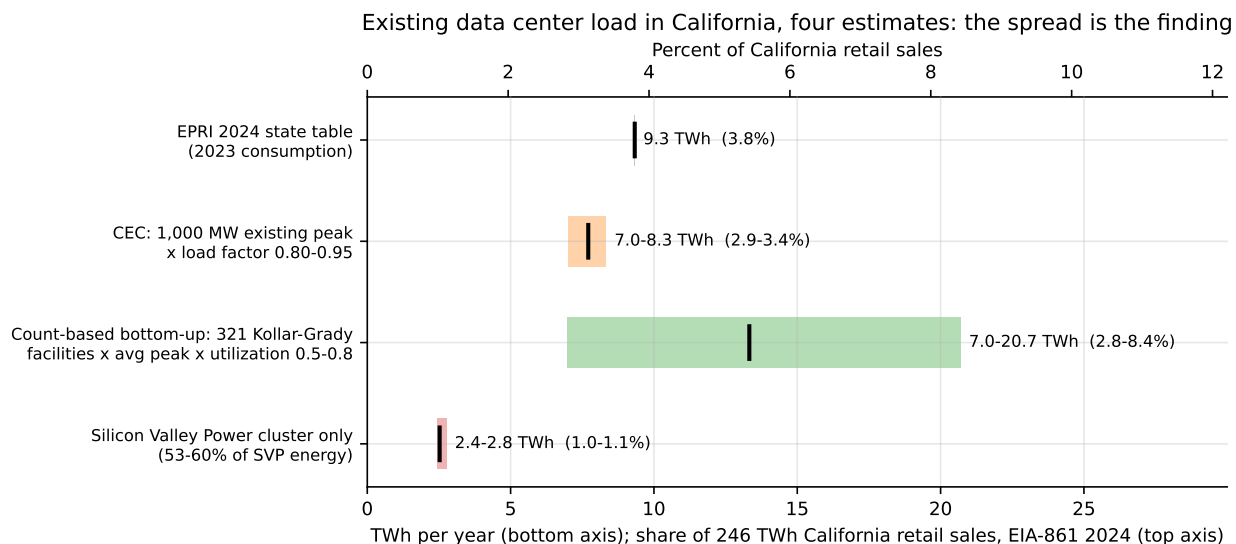


Figure 2.8: Existing data center load in California, four estimates, against 2024 retail sales.

1. **EPRI state table.** EPRI’s 2024 study estimates California data centers consumed 9,331,619 MWh in 2023, 3.70 percent of state consumption, the third-largest state total after Virginia and Texas [Electric Power Research Institute, 2024, pp. 13, 28]. That is 3.8 percent of 2024 retail sales.
2. **CEC existing peak demand.** The CEC’s methodology memo states that “as of December 2025, approximately 1,000 megawatts (MW) of existing data center peak demand operates in California” and that its forecast values “represent incremental growth beyond approximately 1,000 MW of existing data center load” [California Energy Commission, 2026c, pp. 2, 18]. The CEC’s own hourly load-factor profiles for existing facilities range from 75 to 100 percent of annual maximum [California Energy Commission, 2025]; an annual load factor of 0.80 to 0.95 converts the 1,000 MW into 7.0 to 8.3 TWh, or 2.9 to 3.4 percent of retail sales.

3. **Count-based bottom-up.** The Kollar and Grady inventory places 321 of its 2,484 facility points inside California, 102 of them within 12 km of Santa Clara [Kollar and Grady, 2025]; it carries no per-facility power, and Epoch AI’s Frontier Data Centers Hub lists no California site at all [Epoch AI, 2026a]. The estimate therefore multiplies the count by an average peak capacity per facility and a utilization range: the capacity is anchored on Silicon Valley Power, where 58 data centers take 55 percent of a 746 MW system peak, or 7.1 MW per facility, bracketed at minus and plus 30 percent; utilization is 0.50, 0.67 and 0.80, bracketing SVP’s observed 64 to 67 percent of requested capacity. The result, 7.0 to 20.7 TWh (2.8 to 8.4 percent), is the widest of the four because the size and utilization assumptions dominate; it is a bound, not a measurement.
4. **Silicon Valley Power cluster.** The municipal utility of Santa Clara reports 58 data centers, the largest cluster on the West Coast, taking 55 percent of its power use and 51 percent of its sales [Silicon Valley Power, 2026a]; its web page states 53 percent [Silicon Valley Power, 2026b], and city officials cited about 60 percent in 2025 [San José Spotlight, 2025]. SVP’s 2023 fact sheet gives a peak of 669.2 MW and a system load factor of 78.3 percent [Silicon Valley Power, 2024], or about 4.6 TWh a year, so the cluster alone uses 2.4 to 2.8 TWh, about one percent of state retail sales. SVP also reports that its best-performing data center customers reach about 67 percent utilization of requested capacity and that observed utilization fell to about 64 percent in 2023 [Silicon Valley Power, 2025], the empirical origin of the CEC’s 67 percent factor.

The two institutional estimates agree with each other at 7 to 9 TWh, roughly 3 to 4 percent of California’s retail sales, and the bottom-up bound admits values from about 3 to 8 percent. One municipal utility hosts about a third of the institutional estimate. The defensible statement for the rest of this study is that existing data centers use about 3 to 4 percent of the state’s electricity, and that the load is geographically concentrated. Against roughly 1 GW of existing peak, the CEC’s December 2025 energization tiers total 23,277 MW [California Energy Commission, 2026a], a 23-fold ratio that is the California analogue of Finland’s 27x [Manner, 2026] and the subject of Chapter 3.

2.5 Institutional context

The January 2026 forecast decision. The CEC adopted the California Energy Demand 2025–2045 forecast set on January 21, 2026 [California Energy Commission, 2026b]. The set includes three data center scenarios built from the energization tiers. What the joint agencies then chose to do with those scenarios is the institutional fact that matters. The Single Forecast Set Agreement between the CEC, the CPUC and the CAISO, signed in January 2026 and updated on April 15, 2026, directs that CPUC integrated resource planning and CAISO bulk-system transmission studies for the 2026–2027 cycle “will continue to use the 2024 IEPR planning forecast,” that the 2027–2028 cycle will use an IRP portfolio based on the same 2024 forecast, and that system resource adequacy requirements for 2027 “will use the 2025 IEPR planning forecast without known loads” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, pp. 7–8]. Only local transmission reliability studies use the 2025 IEPR local reliability forecast with known loads. The agencies give two reasons: the approach “manages volatility and promotes stability in electric system resource and infrastructure planning currently underway,” and it “allows CEC staff additional time to collect and review historical data, observe the impacts of known loads, data centers, and recent federal policy changes, and refine forecast

approaches as needed” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, p. 8]. In plain terms, the state now tracks data center energization requests in tiers, converts them into forecast scenarios, and then holds the resulting increment out of the forecast that drives generation procurement and bulk transmission for at least two planning cycles. The 2024 IEPR data center mid case, built from the smaller December 2024 tiers, remains the number in use.

Utility pipeline statements. PG&E told the CEC in February 2025 that it had “a large pipeline of applications to interconnect new data centers” with about 1.5 GW in final engineering or construction and more than half expected within four years [Nelli, 2025]. In July 2025 it announced a 10 GW pipeline over ten years, of which 17 projects totaling about 1.5 GW were in final engineering for operation between 2026 and 2030, and estimated that each gigawatt of new data center demand could lower other customers’ bills by one to two percent [Pacific Gas and Electric Company, 2025]. Its second-quarter 2026 earnings release describes “an overall pipeline of over 12 gigawatts” [PG&E Corporation, 2026]. PG&E has, however, filed a request for confidentiality over its project-level data center data in the IEPR docket every month since August 2025 [Pacific Gas and Electric Company, 2026], so its pipeline is public only in aggregate. Southern California Edison took the opposite path. It reported 80 MW of existing data center demand in February 2025 with 200 MW of near-term and 400 MW of mid-term growth [Dean, 2025], and it files a public project-level database. The January 2026 vintage lists 115 entries with 8,298 MW of requested peak: 76 MW under signed agreements, 3,314 MW in applications, 1,773 MW in inquiries, and 3,137 MW already canceled [Southern California Edison, 2026]. The CEC’s own summer 2025 tally put SCE’s requests at 5,828 MW and PG&E’s at 11,668 MW [California Energy Commission, 2025]. The canceled share in SCE’s file, 38 percent of everything ever requested, is the first direct California evidence on attrition and is used in Chapter 5.

CPUC large-load proceedings. The proposal could not confirm a CPUC large-load tariff docket; the record now can. On April 9, 2026 the CPUC opened Rulemaking 26-04-009 on Advanced Electric Rate Design, which states that “in coordination with the Senate Bill 57 assessment, this proceeding will consider rate design issues for data centers and other large load customers” [California Public Utilities Commission, 2026, p. 1]. Senate Bill 57 (Padilla), chaptered on October 11, 2025, requires the CPUC to assess whether costs of serving new data center loads shift to other customers and to report to the Legislature by January 1, 2027 [California Legislature, 2025b]. Senate Bill 886 (Padilla), the California Technology Innovation and Ratepayer Protection Act, was chaptered on September 21, 2026 and requires the CPUC, on or before January 1, 2028, to establish new tariffs or update electric rules for the interconnection and service of participating large customers, and to ensure those tariffs “prevent the creation of stranded costs for, or cost shifts to, nonparticipating customers” [California Legislature, 2026, KQED, 2026]. Assembly Bill 222, which would have required data center energy reporting and efficiency metrics, was held in the Senate Appropriations Committee on August 29, 2025 [California Legislature, 2025a]. On the interconnection side, the CPUC approved an interim PG&E Electric Rule 30 on July 24, 2025 to streamline transmission-level service for large customers that fund the required infrastructure up front [California Public Utilities Commission, 2025]. The sequence is consistent: the state has moved from studying the cost question (SB 57) to mandating a tariff answer (SB 886) within a year, while the forecast that drives procurement still excludes the load the tariff is meant to price.

Chapter 3

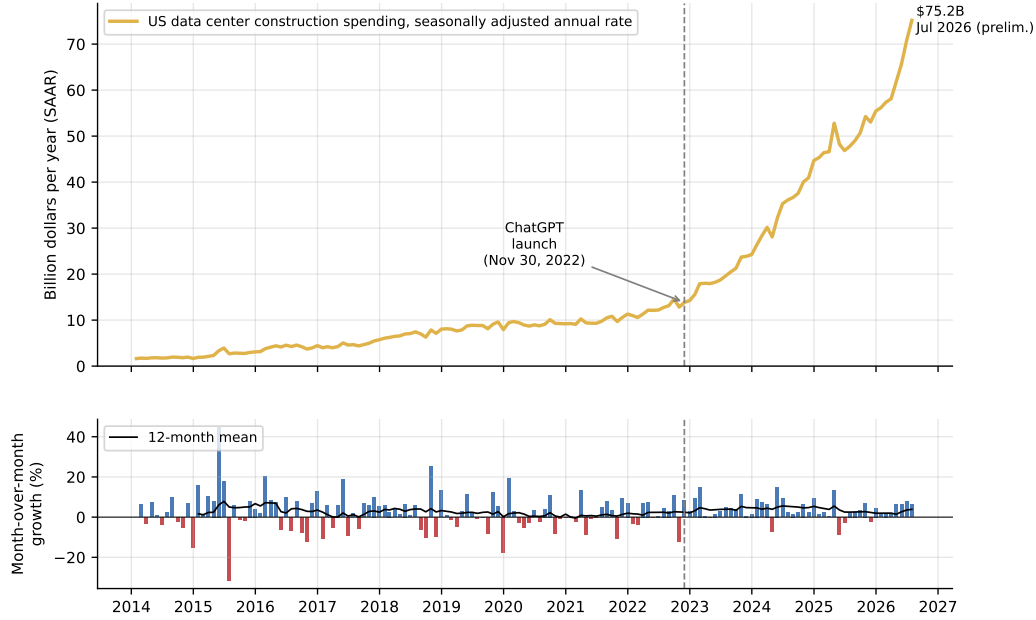
Data Center Growth in California

This chapter answers RQ1: how California’s planned data center capacity varies by stage of development, in absolute terms and relative to the scale of the existing grid. It first reproduces the national construction-spending series that frames public discussion, tests it for structural breaks, and then asks whether California’s own indicators show the same acceleration. It ends with every vintage of the CEC’s energization tiers and the RQ1 table. All code is in `src/ch2_growth.py` and `notebooks/02_data_center_growth.ipynb`.

3.1 The national construction boom, reproduced

The Census Bureau’s Value of Construction Put in Place survey has carried a “data center” line under private nonresidential construction since January 2014 [U.S. Census Bureau, 2026]. Figure 3.1 reproduces the widely circulated chart from the raw seasonally adjusted workbook and adds month-over-month growth. Spending at a seasonally adjusted annual rate rose from \$1.6 billion in January 2014 to \$13.9 billion in November 2022 and \$75.2 billion in July 2026 (preliminary). Data centers were half a percent of private nonresidential construction in 2014, two percent when ChatGPT launched, and ten percent in July 2026. Month-over-month growth averaged 4.1 percent from 2023 through mid-2026 against 1.2 percent in 2019 through 2021. No comparable state-level series exists: the Census publishes this category for the nation only, and the California picture must be built from proxies (Section 3.3).

Data center construction at record highs: US private construction spending, data center category, Jan 2014 to Jul 2026 (Census C30)



Source: U.S. Census Bureau, Value of Construction Put in Place (privatime.xlsx), accessed 2026-09-21; p = preliminary

Figure 3.1: US private data center construction spending, seasonally adjusted annual rate, with month-over-month growth. Source: Census C30, July 2026 preliminary.

3.2 Structural breaks and forecasts

The analysis works on the log of the seasonally adjusted series, so that a linear trend is a constant compound growth rate. Three questions are asked in turn: does a break at the known date change the trend (Chow test, Chow, 1960); where does the data itself place breaks when none is imposed (Bai-Perron, Bai and Perron, 1998, 2003); and what does a time-series model project. Growth rates carry Newey-West standard errors; the Bai-Perron search uses 15 percent trimming (a minimum segment of 23 months), chooses the number of breaks by BIC and by the LWZ criterion, reports $\text{supF}(0|1)$ with a moving-block residual bootstrap p -value under the no-break null, and gives the single-break date a residual-bootstrap 90 percent interval. The `ruptures` dynamic-programming search [Truong et al., 2020] is run as an independent check.

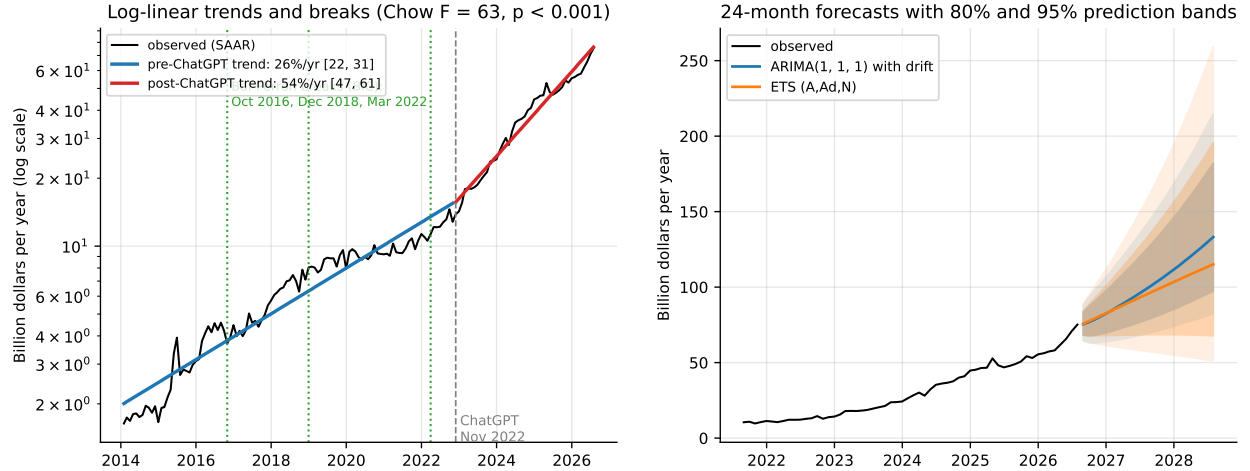


Figure 3.2: Left: log-linear trends before and after December 2022 with Bai-Perron break dates. Right: 24-month ARIMA and ETS forecasts with 80 and 95 percent prediction bands.

Table 3.1: Bai-Perron global search on the log Census series: break dates, sum of squared residuals, information criteria and supF statistics for $m = 0$ to 3 breaks.

Breaks m	Break dates (first month of new regime)	SSR	BIC	LWZ	supF(0 m)
0	—	5.967	-3.165	-3.101	—
1	2020-11	2.205	-4.061	-3.900	125.4
2	2018-12, 2022-03	1.375	-4.433	-4.176	121.1
3	2016-10, 2018-12, 2022-03	1.091	-4.565	-4.210	106.5

Table 3.2: Compound growth by segment of the BIC-selected partition (Newey-West 95 percent intervals).

Segment (BIC partition)	Months	Compound growth (%/yr)	95% interval
2014-01 to 2016-09	33	53	[45, 61]
2016-10 to 2018-11	26	42	[37, 47]
2018-12 to 2022-02	39	9	[7, 10]
2022-03 to 2026-07	53	55	[50, 61]

The known-date test is unambiguous. A break in intercept and slope at December 2022 gives a Chow F of 63 ($p < 0.001$) and the HAC-robust Wald test agrees ($p < 0.001$). Compound growth was 26 percent a year (95 percent interval 22 to 31) before the launch and 54 percent a year (47 to 61) after it. Letting the data locate the breaks refines the timing (Tables 3.1 and 3.2). Both criteria choose three breaks, in October 2016, December 2018 and March 2022, so the current regime began eight months before ChatGPT, in the hyperscaler capital-spending ramp of early 2022, and the segment from March 2022 to July 2026 grows at 55 percent a year (50 to 61). The single-break supF of 125 has a bootstrap p -value of 0.01, and `ruptures` reproduces every break date within one month. The forecasts in Table 3.3 continue the current trend: an ARIMA(1,1,1) with drift and a damped-trend ETS both put July 2027 near \$95 to \$100 billion and July 2028 near \$115 to \$135 billion, with 95 percent bands from roughly \$50 to \$260 billion by then. The bands, not the

point forecasts, are the finding: a series growing at 55 percent a year with a one-third chance of a further break is not something a planner should extrapolate.

Table 3.3: Forecasts of US data center construction spending, 6, 12 and 24 months ahead.

Month	ARIMA(1, 1, 1) with drift			ETS(A,Ad,N)		
	Mean	80% band	95% band	Mean	80% band	95% band
Jan 2027	84.5	[70.7, 101.0]	[64.3, 111.0]	84.4	[68.0, 104.7]	[60.6, 117.4]
Jul 2027	98.3	[78.0, 124.0]	[68.9, 140.2]	94.8	[68.4, 131.3]	[57.6, 156.0]
Jul 2028	133.2	[97.4, 182.1]	[82.5, 214.9]	115.1	[67.8, 195.6]	[51.2, 258.9]

Billion dollars per year, seasonally adjusted annual rate.

3.3 California proxies

Three California indicators are subjected to the same battery, with the known break set to the first period after November 2022 (Figure 3.3, Table 3.4).

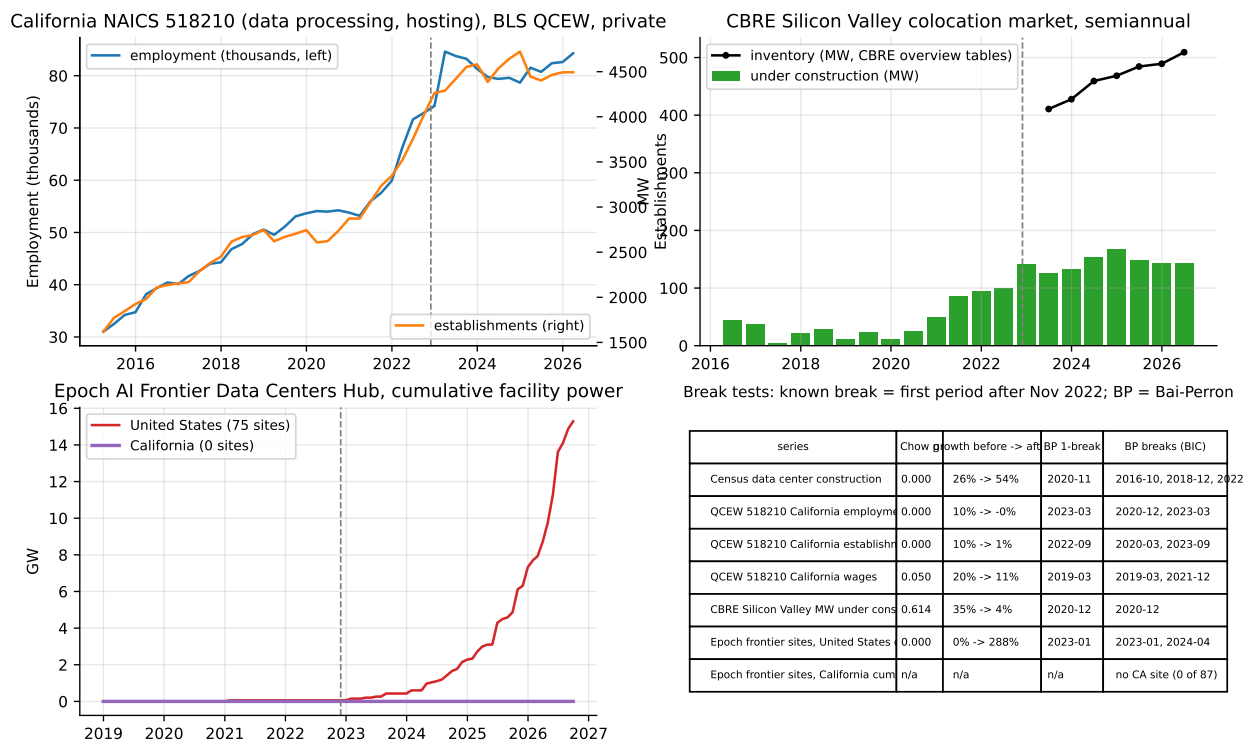


Figure 3.3: California proxies for data center growth: BLS QCEW NAICS 518210 (top left), CBRE Silicon Valley construction and inventory (top right), Epoch AI frontier sites (bottom left), and the break-test summary (bottom right).

Table 3.4: Break tests on the Census series and the California proxies. Chow: known break at the first period after November 2022. BP: Bai-Perron single break with residual-bootstrap 90 percent interval, BIC-selected breaks, and the bootstrap p -value of $\sup F(0|1)$.

Series	n	Chow F	p	HAC p	Growth before (%/yr)	Growth after (%/yr)	BP break (1)	90% interval	BP breaks (BIC)	boot. p
Census data center construction (monthly, SAAR)	151	63.0	0.000	0.000	26 [22, 31]	54 [47, 61]	2020-11	–	2016-10, 2018-12, 2022-03	0.010
QCEW 518210 California employment (quarterly)	45	23.9	0.000	0.000	10 [8, 11]	-0 [-2, 2]	2023-03	–	2020-12, 2023-03	0.065
QCEW 518210 California establishments (quarterly)	45	13.8	0.000	0.000	10 [8, 12]	1 [-1, 2]	2022-09	–	2020-03, 2023-09	0.095
QCEW 518210 California wages (quarterly)	45	3.2	0.050	0.000	20 [18, 23]	11 [9, 13]	2019-03	–	2019-03, 2021-12	0.005
CBRE Silicon Valley MW under construction (semiannual)	21	0.5	0.614	0.000	35 [10, 66]	4 [-2, 10]	2020-12	–	2020-12	0.140
Epoch frontier sites, United States cumulative MW (monthly)	69	176.1	0.000	0.000	0 [-0, 0]	288 [250, 331]	2023-01	–	2023-01, 2024-04	0.005
Epoch frontier sites, California cumulative MW	0	no California site in the Epoch hub (0 of 87); series is identically zero, no test possible								

Employment and establishments. The Quarterly Census of Employment and Wages counts private establishments and jobs in NAICS 518210, data processing, hosting and related services, by quarter [U.S. Bureau of Labor Statistics, 2026]. California employment in the industry rose from 31,000 in 2015 to 74,000 by the end of 2022, then flattened: 84,000 in 2026 Q1 after a plateau through 2024 and 2025. The Chow test at 2023 Q1 rejects a common trend ($F = 24$, $p < 0.001$), but the break is a deceleration, from 10 percent a year to roughly zero. Establishments show the same shape (10 percent a year to 0.5 percent), and Bai-Perron places their break between 2022 Q3 and 2023 Q2. Wages decelerated from 20 to 11 percent a year with breaks in 2019 and 2021 that reflect the industry’s earlier expansion. The county detail is in the notebook; Santa Clara, Los Angeles and San Francisco account for most of the state total.

MW under construction. CBRE’s semiannual market chapters give Silicon Valley colocation capacity under construction from H1 2016 and inventory from H1 2023 [CBRE Research, 2026]. Construction rose from 44 MW in H1 2016 to 142 MW by H2 2022 and has since ranged between 125 and 168 MW, with inventory creeping from 411 to 509 MW. The Chow test finds no break at H1 2023 ($p = 0.61$); Bai-Perron puts the only break in H2 2020, when the pipeline stepped up from tens of megawatts to over a hundred, and growth has been 4 percent a year since. Cushman & Wakefield’s tables are form-gated and JLL’s report gives a single-period inventory, so the digitized series rests on CBRE; Southern California is available only from H1 2024 and is reported in the notebook.

Frontier facilities. Epoch AI’s hub tracks 87 large AI data centers from satellite imagery and permits, 75 of them in the United States, with per-site power timelines [Epoch AI, 2026b]. Cumulative US power in these sites went from under 0.5 GW at the end of 2023 to 15.3 GW by September 2026, with a break in January 2023. Not one of the 87 sites is in California, so the California series is identically zero and no test is possible.

The three proxies agree on a finding that the national series hides: after 2022 California’s data center industry did not accelerate in jobs, establishments or built colocation capacity, and the largest AI campuses were sited elsewhere. The growth that California records is in requests to connect, which is the subject of the next section, and the divergence between requests and built capacity is exactly what the tiers are meant to measure.

3.4 Every vintage of the energization tiers

Table 3.5 and Figure 3.4 assemble every published vintage of the CEC’s data center energization requests. The December 2024 data behind the 2024 IEPR Update are known only as PG&E’s and SCE’s agreements plus applications, 5,808 and 963 MW, because inquiries were not published [California Energy Commission, 2025]. The summer 2025 data behind the preliminary 2025 forecast are published as totals by utility, 21,756 MW across seven utilities, with the tier split shown graphically [California Energy Commission, 2025]. The December 2025 data behind the adopted forecast are the first complete vintage: 5,086 MW signed, 9,587 MW in application and 8,604 MW in inquiry, 23,277 MW in all, with PG&E accounting for 14,747 MW and SCE for 4,623 MW [California Energy Commission, 2026c,a]. The August 2026 workshop restated the December 2025 figures and showed June 2026 known-load data only as charts [Smith, 2026]. SCE’s public project database provides the only utility-level vintages with cancellations: between its August 2025 and January 2026 versions, inquiries fell from 5,934 to 1,772 MW as about 3,100 MW moved into the application stage, and canceled requests rose from 709 to 3,137 MW [Southern California Edison, 2026]. Thirty-eight percent of everything SCE had ever logged was canceled within five months, the first direct California evidence on attrition.

Table 3.5: Vintages of the CEC energization tiers found in the record (MW of requested capacity).

Vintage	Signed	Applications	Inquiries	Agr.+appl. only	Unsplit total	Total active (MW)
Dec 2024 (2024 IEPR Update)	0	0	0	6,771	0	6,771
Summer 2025 (2025 IEPR preliminary)	0	0	0	0	21,756	21,756
Dec 2025 (2025 IEPR final)	5,086	9,587	8,605	0	0	23,278
Aug 2026 workshop (restates Dec 2025)	0	0	0	0	23,278	23,278
SCE database Aug 29 2025 (SCE only)	51	216	5,934	–	canceled 709	6,201
SCE database Jan 29 2026 (SCE only)	76	3,313	1,772	–	canceled 3,137	5,162

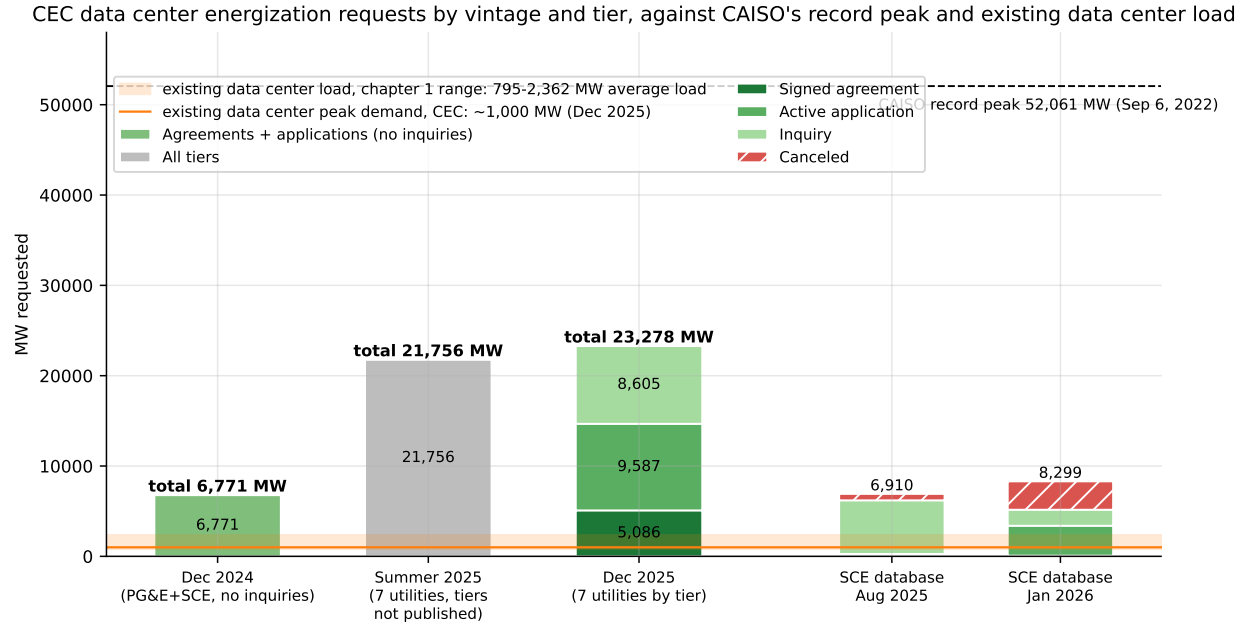


Figure 3.4: Energization requests by vintage and tier against CAISO's record peak and the existing data center load from Chapter 2.

The sums and ratios are stated explicitly. The three December 2025 tiers total 23,277 MW. Against CAISO's all-time peak of 52,061 MW on September 6, 2022 [California Independent System Operator, 2026a], that is 45 percent (46 percent against the hourly EIA-930 peak of 51,104 MW). Against the roughly 1,000 MW of existing data center peak demand that the CEC reports for December 2025, it is 23 times; against the chapter 1 range of existing average load, 795 to 2,362 MW, it is 10 to 29 times. Finland's ratio of registered to installed capacity was 27 [Manner, 2026]; California's is of the same order.

3.5 RQ1: planned capacity by stage

Table 3.6: RQ1: California's planned data center capacity by stage (December 2025) relative to the grid, to existing data center load, and to the adopted forecast's growth to 2030.

Stage (Dec 2025)	MW	% of CAISO record peak	% of 2025 peak	× existing DC peak	× existing DC avg. load	% of IEPR peak growth	% of IEPR growth at 67% util.
Signed agreements	5,086	9.8	11.6	5.1x	2.2-6.4x	127	85
Active applications	9,587	18.4	21.9	9.6x	4.1-12.1x	239	160
Inquiries	8,604	16.5	19.6	8.6x	3.6-10.8x	214	143
Total, three tiers	23,277	44.7	53.1	23.3x	9.9-29.3x	579	388

Denominators: CAISO record peak 52,061 MW (Sept. 6, 2022); 2025 hourly peak 43,860 MW; existing data center peak ≈1,000 MW (CEC); existing average load 795–2,362 MW (chapter 1 range); IEPR planning-scenario CAISO managed net peak growth 2025–2030 4,019 MW.

Table 3.6 answers RQ1. The denominators come from the sources of the preceding chapters: the CAISO record and 2025 peaks, the CEC's existing-load figure and the chapter 1 range, and the adopted 2025 IEPR planning forecast, whose hourly CAISO forecast gives a managed net peak of

46,479 MW in 2025 and 50,498 MW in 2030, a growth of 4,019 MW, and whose statewide energy to serve load grows from 263 to 298 TWh [California Energy Commission, 2026d]. Signed agreements alone, 5,086 MW, exceed the forecast’s entire five-year peak growth (127 percent of it); the three tiers together are 5.8 times that growth, 3.9 times it even at the CEC’s 67 percent utilization factor, and their energy equivalent at that utilization and a 0.88 load factor, about 120 TWh a year, is 3.4 times the forecast’s statewide energy growth to 2030. The forecast is not blind to data centers: its planning scenario carries a data center contribution at the CAISO peak that rises from 96 MW in 2025 to 1,743 MW in 2030 (4,377 MW in the local-reliability scenario), and data-center-only deliveries of 12.1 TWh in 2030, roughly a third of projected statewide energy growth. What the forecast does is discount the requests, by the confidence levels and utilization of Chapter 2, to about a tenth of their face value at the peak by 2030. Whether that discount is right is the question the remaining chapters take up with the supply side, the attrition evidence and the adequacy framing.

Chapter 4

Energy Production Growth in California

Chapter 5

Putting It Together

Chapter 6

Discussion

Bibliography

- Jushan Bai and Pierre Perron. Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78, 1998.
- Jushan Bai and Pierre Perron. Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1):1–22, 2003.
- California Energy Commission. 2025 iepr: Preliminary data center forecast (iepr commissioner workshop deck, october 30, 2025), 2025. manifest: cec_prelim_dc_forecast_2025; docket 25-IEPR-03 TN 267165.
- California Energy Commission. Presentation to the assembly utilities and energy and privacy and consumer protection committees joint oversight hearing on the energy impacts of artificial intelligence, january 28, 2026, 2026a. manifest: cec_assembly_hearing_2026_01_28; energization tiers on slide 7.
- California Energy Commission. Resolution adopting the california energy demand forecast, 2025–2045 (business meeting item 6, january 21, 2026), 2026b. manifest: cec_ced_2025_2045_report, cec_ced_2025_2045_resolution.
- California Energy Commission. Data center methodology memo: Supporting document for the 2025 iepr forecast. Technical report, April 2026c. manifest: cec_dc_methodology_memo_2026.
- California Energy Commission. Ced 2025 planning forecast lse and baa tables (forms 1.1c, 1.5a, 1.5b), ced 2025 peak forecast (hourly-forecast annual peaks by scenario), and form 1.1c data center allocations, 2026d. manifest: cec_tn268727, cec_tn268124, cec_tn268824; docket 25-IEPR-03.
- California Energy Commission, California Public Utilities Commission and California Independent System Operator. 2025 iepr forecast: Single forecast set agreement, updated april 15, 2026 (supersedes the january 2026 agreement, tn 268288), 2026. manifest: cec_tn269506, cec_tn268288; docket 25-IEPR-03.
- California Independent System Operator. Key statistics, august 2026: all-time peak 52,061 mw on september 6, 2022. *CAISO monthly key statistics*, 2026a. manifest: caiso_key_statistics_2026_08.
- California Independent System Operator. Oasis day-ahead market locational marginal prices (prc_lmp, dam), dlap_pgae-apnd, dlap_sce-apnd, dlap_sdge-apnd and np15/sp15/zp26 trading hubs, july 2023 – september 2026, 2026b. manifest: caiso_dam_lmp_monthly_dlap, caiso_dam_lmp_monthly_hubs; accessed 2026-09-21.

California Independent System Operator. Today’s outlook historical five-minute data: Co₂ emissions by source, demand, fuel mix, april 2018 – september 2026, 2026c. manifest: caiso_outlook_co2, caiso_outlook_demand, caiso_outlook_fuelsource; accessed 2026-09-21.

California Legislature. Assembly bill 222 (bauer-kahan), data centers: power usage effectiveness: cost shifts. held in senate appropriations, august 29, 2025, 2025a. manifest: ca_ab222_2025_status.

California Legislature. Senate bill 57 (padilla), electrical corporations: data centers: rates. chaptered october 11, 2025, statutes of 2025, chapter 647, 2025b. manifest: ca_sb57_2025_status, ca_sb57_2025_text.

California Legislature. Senate bill 886 (padilla), california technology innovation and ratepayer protection act. chaptered september 21, 2026, 2026. manifest: ca_sb886_2026_status, ca_sb886_2026_text.

California Public Utilities Commission. Cpuc streamlines electric grid connections for high-energy users like data centers and ev chargers (news release, july 24, 2025), 2025. manifest: cpuc_news_streamlined_connections.

California Public Utilities Commission. Order instituting rulemaking on california advanced electric rate design, rulemaking 26-04-009, April 2026. manifest: cpuc_oir_rate_design_2026_04; issued April 10, 2026.

CBRE Research. North america data center trends, h1 2023 – h1 2026 editions: Silicon valley market chapter history and primary/secondary market tables (infogram chart data), 2026. manifest: cbre_h1_2026_silicon_valley_infogram_fig2 and cbre_*_overview_infogram_*; assembled in data/processed/cbre_california_market_series.csv.

Gregory C. Chow. Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3):591–605, 1960.

Elliot Dean. Sce data center forecast (iepr commissioner workshop on california’s economic outlook, february 26, 2025), 2025. manifest: cec_tn261975; docket 25-IEPR-03 TN 261975.

Electric Power Research Institute. Powering intelligence: Analyzing artificial intelligence and data center energy consumption. Technical Report 3002028905, May 2024. manifest: epri_powering_intelligence_2024.

Epoch AI. Frontier data centers hub (data_centers.csv, vintage of september 21, 2026), 2026a. manifest: epoch_data_centers.

Epoch AI. Frontier data centers hub: data_center_timelines.csv (construction status and power by date, vintage of september 21, 2026), 2026b. manifest: epoch_data_center_timelines.

Abbey Kollar and Caitlin Grady. The relationship between data centers and the climate is a systems challenge: a spatial analysis of united states data centers. *Environmental Research Communications*, 7(11), 2025. Dataset: Zenodo 10.5281/zenodo.17372375; manifest: kollar_grady_2025_zenodo.

KQED. Newsom signs new restrictions on data center development, September 2026. manifest: kqed_newsom_signs_data_center_bills_2026.

Jukka Manner. The data centre boom in finland and its grid implications. Technical report, IRTF SUSTAIN workshop, Passau, July 2026.

Daniel Nelli. Pg&e data center pipeline (iepr commissioner workshop on california’s economic outlook, february 26, 2025), 2025. manifest: cec_tn261964; docket 25-IEPR-03 TN 261964.

Pacific Gas and Electric Company. Pg&e data center demand pipeline swells to 10 gigawatts with potential to unlock billions in benefits for california (press release, july 31, 2025), 2025. manifest: pge_press_2025_07_31_pipeline_10gw.

Pacific Gas and Electric Company. Data center requests for confidentiality filed in docket 25-iepr-03 and 26-iepr-03, august 2025 – september 2026, 2026. manifest: cec_tn265852, cec_tn267896, cec_tn268283, cec_tn268596, cec_tn272858.

PG&E Corporation. Form 8-k, second-quarter 2026 earnings press release, 2026. manifest: pge_8k_q2_2026.

San José Spotlight. Santa clara data centers hit max energy capacity, 2025. manifest: sjspotlight_santa_clara_capacity_2025.

Silicon Valley Power. Utility fact sheets 2017–2023, 2024. manifest: svp_fact_sheet_2017 ... svp_fact_sheet_2023 (Wayback Machine captures).

Silicon Valley Power. Presentation to the cec (2025): planning, load growth and data center utilization, 2025. manifest: cec_tn264908.

Silicon Valley Power. Presentation to the assembly joint oversight hearing on the energy impacts of artificial intelligence, january 28, 2026, 2026a. manifest: svp_assembly_hearing_2026_01_28.

Silicon Valley Power. Data centers in santa clara (web page, wayback machine capture of july 9, 2026), 2026b. manifest: svp_data_centers_page_wayback.

Jeremy Smith. Data center demand forecast (iepr commissioner workshop on energy demand forecast inputs and assumptions, august 20, 2026), 2026. manifest: cec_tn272026; docket 26-IEPR-03 TN 272026.

Southern California Edison. See iepr25 data center database, public version of january 29, 2026, 2026. manifest: cec_tn268459; docket 25-IEPR-03 TN 268459; earlier vintage TN 266008.

Charles Truong, Laurent Oudre, and Nicolas Vayatis. Selective review of offline change point detection methods. *Signal Processing*, 167:107299, 2020. ruptures Python package, used as a cross-check.

U.S. Bureau of Labor Statistics. Quarterly census of employment and wages, naics 518210 (data processing, hosting and related services), private ownership, california and counties, 2015 q1 – 2026 q1, 2026. manifest: qcew_518210_2015q1 ... qcew_518210_2026q1.

U.S. Census Bureau. Value of construction put in place (c30): private construction, data center category, monthly, seasonally adjusted annual rate and not seasonally adjusted, january 2014 – july 2026, 2026. manifest: census_c30_privsatime, census_c30_privtime, census_c30_privsa; accessed 2026-09-21; July 2026 preliminary.

- U.S. Energy Information Administration. Annual electric power industry report (form eia-861), sales to ultimate customers, 2019–2024, 2025. manifest: eia861_2019 ... eia861_2024; accessed 2026-09-21.
- U.S. Energy Information Administration. Hourly electric grid monitor (form eia-930), balancing authority ciso, six-month bulk files 2019–2025, 2026a. manifest: eia930_balance_2019_jan_jun ... eia930_balance_2025_jul_dec, eia930_reference_tables; accessed 2026-09-21.
- U.S. Energy Information Administration. Wholesale electricity market data from the intercontinental exchange: daily day-ahead block prices at np15 and sp15, 2015–2026, 2026b. manifest: eia_ice_wholesale_2015 ... eia_ice_wholesale_2026.
- U.S. Environmental Protection Agency. Emissions & generation resource integrated database (egrid) 2023, data file rev1, subregion camx, 2025. manifest: epa_egrid2023_rev1.